

Knowledge

ECON 403/503



Knowledge Spaces

- Ω = set of states of the world (finite)
 $\omega \in \Omega$, ω = incorporates all resolution of uncertainty pertaining to the situation.

Knowing ω means there is no uncertainty

- N set of players = $\{1, \dots, n\}$
- A player's type t_i will represent the type of information a player has.

Definition: A Knowledge space is the following tuple:

$$(N, \Omega, \{T_i\}_{i \in N}, (\tau_i)_{i \in N}) \text{ where}$$

- N is the set of players
- Ω is the set of states of the world
- T_i is the set of player types (private to player i)
- $\tau_i: \Omega \rightarrow T_i$ is player's i private information function (type)

Example: $N = \{1, 2\}$, $\Omega =$ set of all possible outcomes of rolling two dice.

Suppose P_1 sees only the first die
 P_2 sees only the sum

$$\Omega = \{(k, l) : k = 1, \dots, 6, l = 1, \dots, 6\}$$

$$T_1 = \{1, \dots, 6\}, T_2 = \{2, \dots, 12\}$$

$$\tau_1(k, l) = k$$

$$\tau_2(k, l) = k + l$$

Example: The "hats" problem.

$$N \text{ players, } \Omega = \{r, b\}^n, \quad T_i = \{r, b\}^{n-1}, \quad \tau_i(\omega) = \omega_{-i}$$

$$\omega = (\omega_i, \omega_{-i})$$

$$N = \{1, 2, 3\}, \quad \omega = \{b, r, r\}$$

$$\tau_1(\omega) = \{r, r\}, \quad \tau_2(\omega) = \{b, r\}, \quad \tau_3(\omega) = \{b, r\}$$

We are going to define the information partition function of player i

$$P_i: \Omega \rightarrow 2^\Omega$$

is defined by:

2^Ω denotes the power set of Ω ,
or the set of all subsets of Ω
 $2^\Omega = \{A \subseteq \Omega\}$

$$\begin{aligned} P_i(\omega) &= \{\omega' : \tau_i(\omega') = \tau_i(\omega)\} \\ &= \tau_i^{-1}(\tau_i(\omega)) \end{aligned}$$

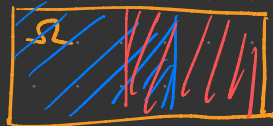
Some simple properties of $P_i : P_i(\omega) = \{\omega' : \tau_i(\omega') = \tau_i(\omega)\}$

- $\omega \in P_i(\omega)$
- $\{P_i(\omega)\}_{\omega \in \Omega}$ constitutes a partition of Ω

Defn: A partition of Ω is a family $\mathcal{P}$ of pairwise-disjoint non-empty subsets of Ω whose union includes Ω .

$$P1: \bigcup_{P \in \mathcal{P}} P = \Omega$$

$$P2: A, B \in \mathcal{P}, A \cap B \neq \emptyset \Rightarrow A = B$$



← NOT a partition.

Back to the Hats example: $N = \{1, 2\}$

$$\Omega = \{(r, r), (r, b), (b, r), (b, b)\}$$

$$\begin{array}{l} P_1(r, r) = \{(r, r), (b, r)\} = P_1(b, r) \\ P_1(r, b) = P_1(b, b) = \{(r, b), (b, b)\} \end{array} \quad \left| \quad \begin{array}{l} P_2(r, r) = P_2(r, b) = \{(r, r), (r, b)\} \\ P_2(b, r) = P_2(b, b) = \{(b, r), (b, b)\} \end{array} \right.$$

A Knowledge space can also be given as $(N, \Omega, \{P_i\}_{i \in N})$

$$\begin{array}{l} A = \{\text{At least one red hat}\} \\ = \{(r, r), (r, b), (b, r)\} \end{array} \quad \left. \vphantom{\begin{array}{l} A \\ = \end{array}} \right\} \begin{array}{l} K_1 A = A \\ K_2 A = A \end{array}$$

$$\begin{array}{l} B = \{\text{There are two red hats}\} \\ = \{(r, r)\} \end{array}$$

Knowledge

Let $(N, \Omega, \{T_i\}_{i \in N}, \{\tau_i\}_{i \in N})$ be a knowledge space

Let $A \in 2^\Omega$ be an event.

We will say that player i knows A at ω if

$P_i(\omega) \subseteq A$, or equivalently if $\tau_i(\omega') = \tau_i(\omega)$ implies that $\omega' \in A$.

Given an event $A \in 2^\Omega$, let $K_i A$ be the set of states of the world in which i knows A :

$$K_i A = \{\omega : P_i(\omega) \subseteq A\}$$

Equivalently,

$$K_i A = \bigcup \{P_i(\omega) : P_i(\omega) \subseteq A\}$$

